GEOGRAPHY

PHYSICAL GEOGRAPHY




CLIMATOLOGY

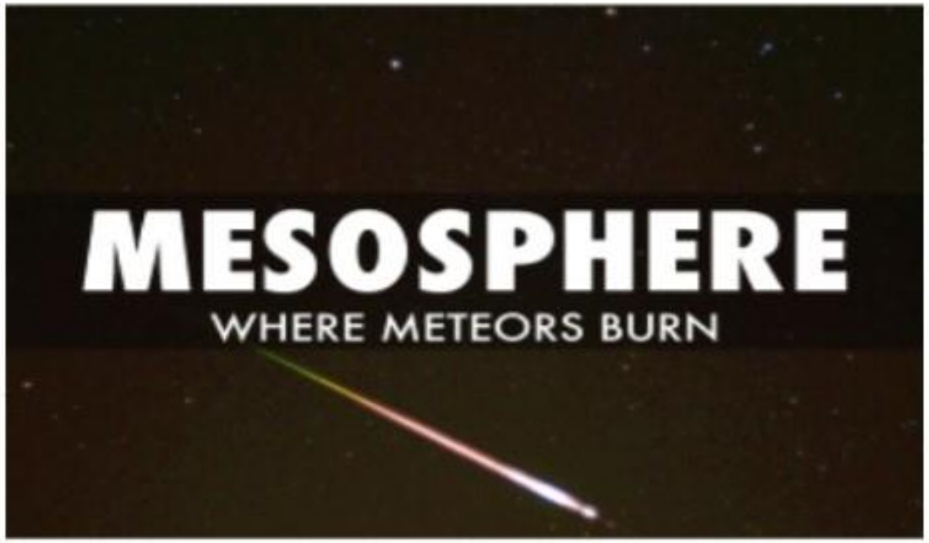
MODULE – 7

STRUCTURE OF ATMOSPHERE

- MESOSPHERE
- THERMOSPHERE

STRUCTURE OF THE ATMOSPHERE

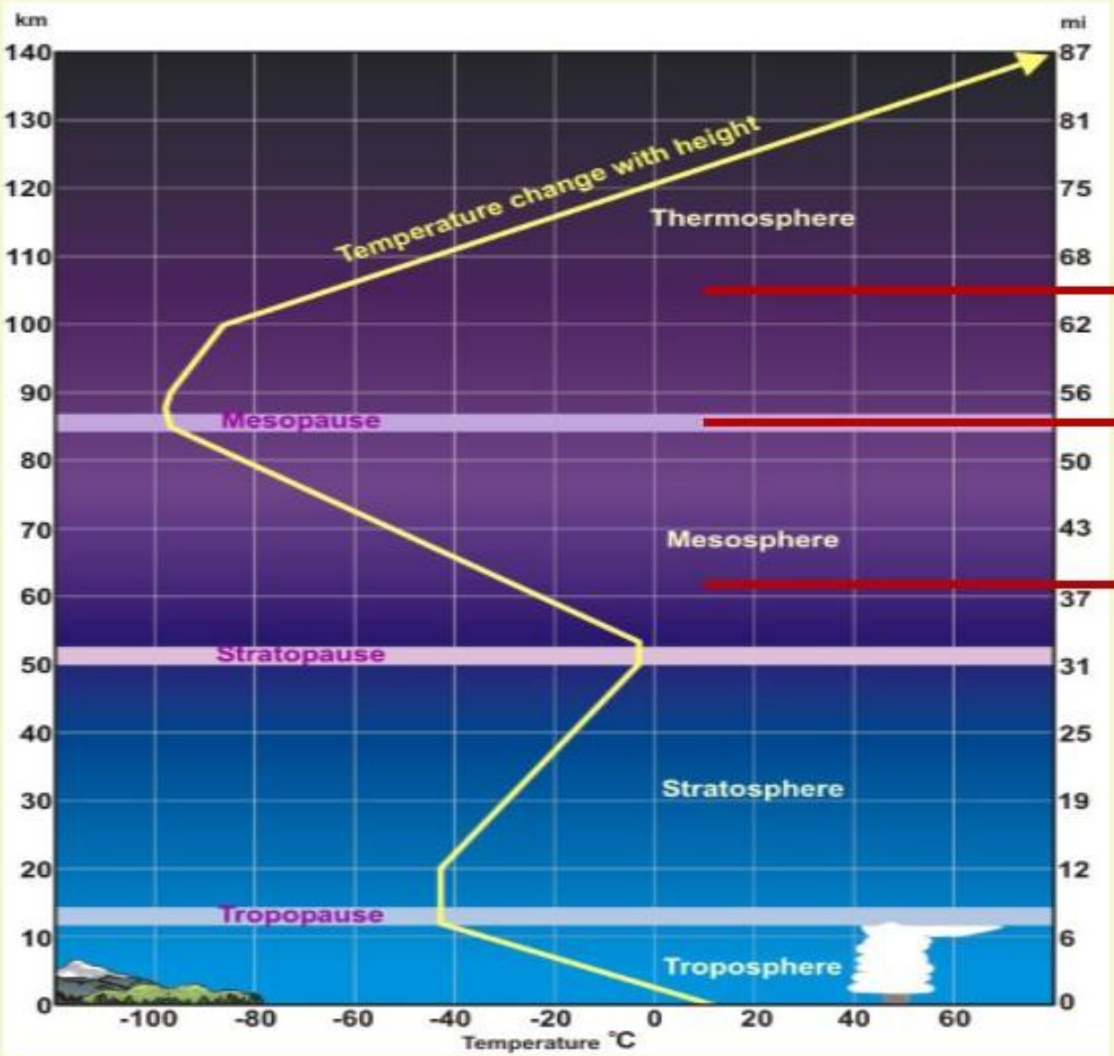
3 MESOSPHERE



1. Mesosphere extends between 50 km and 80km.
2. Temperature again decreases with increasing height.
3. Uppermost limit is known as mesopause.
4. Above mesopause, temperature increases with increasing height.

STRUCTURE OF THE ATMOSPHERE

TEMPERATURE PROFILE IN ATMOSPHERE LAYERS



Temp increases with height from here onwards

Isothermal layer

Temp decreases with height

Learning Space

STRUCTURE OF THE ATMOSPHERE

3 MESOSPHERE

POLAR MESOSPHERIC CLOUDS

1. PMCs are otherwise known as noctilucent clouds or night shining clouds.
2. Noctilucent clouds form in the highest reaches of the atmosphere – the mesosphere – as much as 80 km above the Earth's surface.
3. They can only form when temperatures are incredibly low so that water available there can transform into ice crystals.

STRUCTURE OF THE ATMOSPHERE

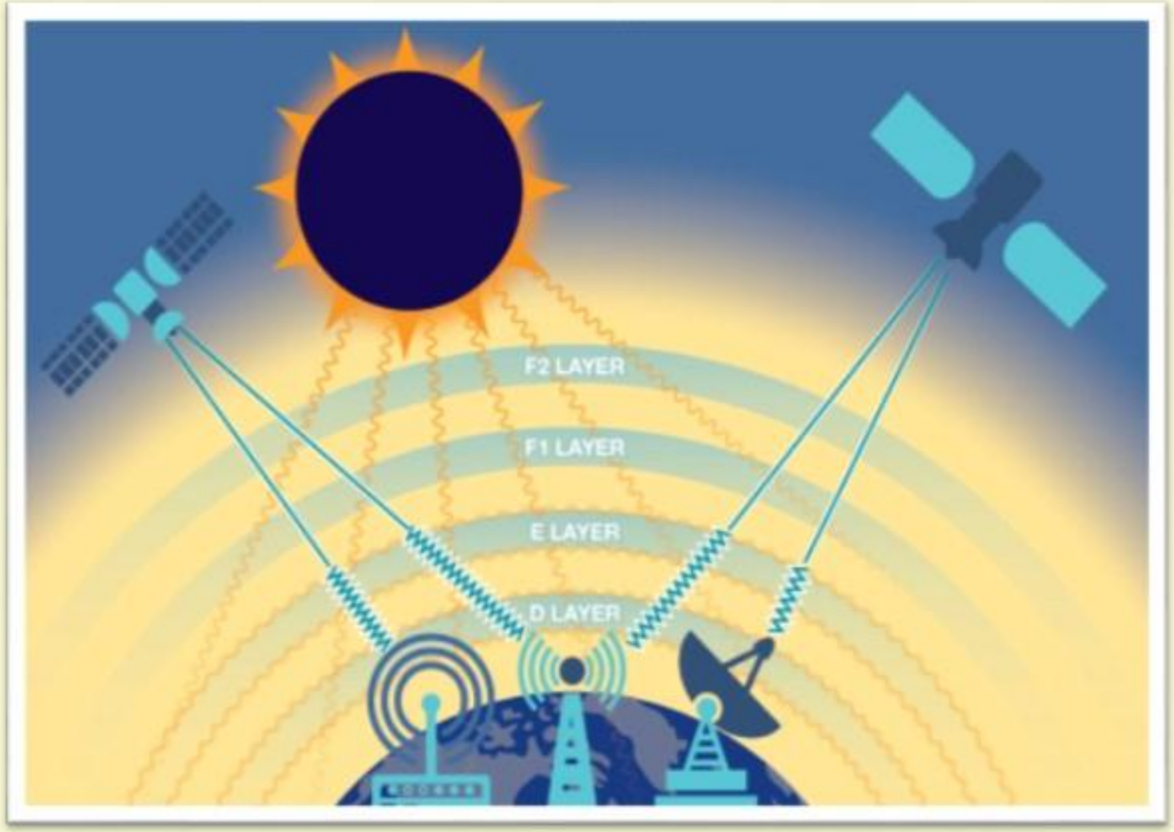
4 THERMOSPHERE

1. Thermosphere extends between 80km and 640km.
2. Temperature in its upper limit becomes upto 1700 degree Celsius. However this temperature cannot be measured by ordinary thermometer because the gases become very light due to extremely low density.
3. Ionosphere is an important part of thermosphere.

STRUCTURE OF THE ATMOSPHERE

4 THERMOSPHERE

IONOSPHERE



1. The ionosphere is the ionized part of Earth's upper atmosphere.
2. The ionosphere is ionized by UV component of solar radiation.
3. It influences radio propagation to distant places on the Earth.

STRUCTURE OF THE ATMOSPHERE

4 THERMOSPHERE

IONOSPHERE

There are a number of ionic layers in ionosphere- D layer, E layer, F layer.

D LAYER

1. D layer (between the height of 60 km-99 km) reflects the signals of low frequency radio waves.
2. This layer disappears with the sunset because it is associated with solar radiation.

E LAYER

1. E layer, also known as **Kennelly-Heaviside layer** is confined within the height between 99 km -130 km.
2. This layer is produced due to interaction of solar ultraviolet thus it also disappears with the sunset.
3. This layer reflects the medium and high frequency radio waves back to the earth.

F LAYER

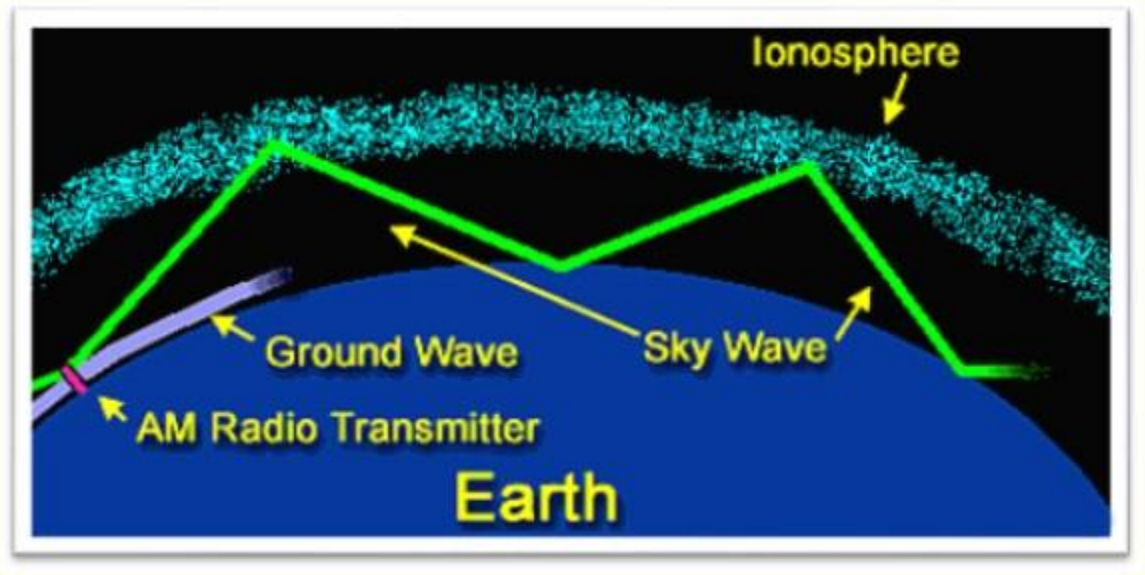
1. This layer is also known as the **Appleton layer.**
2. These layers reflect medium and high frequency radio waves back to the earth.
3. The F region exists during both daytime and nighttime. During the day it is ionized by solar radiation, during the night by cosmic rays.

STRUCTURE OF THE ATMOSPHERE

4 THERMOSPHERE

IONOSPHERE

Why are radio transmissions weaker at night than during the day?



1 Ionization depends primarily on the Sun and its activity. Hence the amount of ionization in the ionosphere varies greatly with the amount of radiation received from the Sun. Thus there is a diurnal effect and a seasonal effect.

2 At night, of course, much less light is entering the atmosphere, which means that less ionization is occurring and fewer electrons are available for radio waves to bounce off. The result is that radio transmissions are weaker.

STRUCTURE OF THE ATMOSPHERE

4 THERMOSPHERE

IONOSPHERE

A layer in the Earth's atmosphere called Ionosphere facilitates radio communication. Why?

1. The presence of ozone caused the reflection of radio waves to Earth.
2. Radio waves have a very long wavelength.

Which of the statements given above is/are correct? (UPSC Prelims 2011)

A. 1 only

B. 2 only

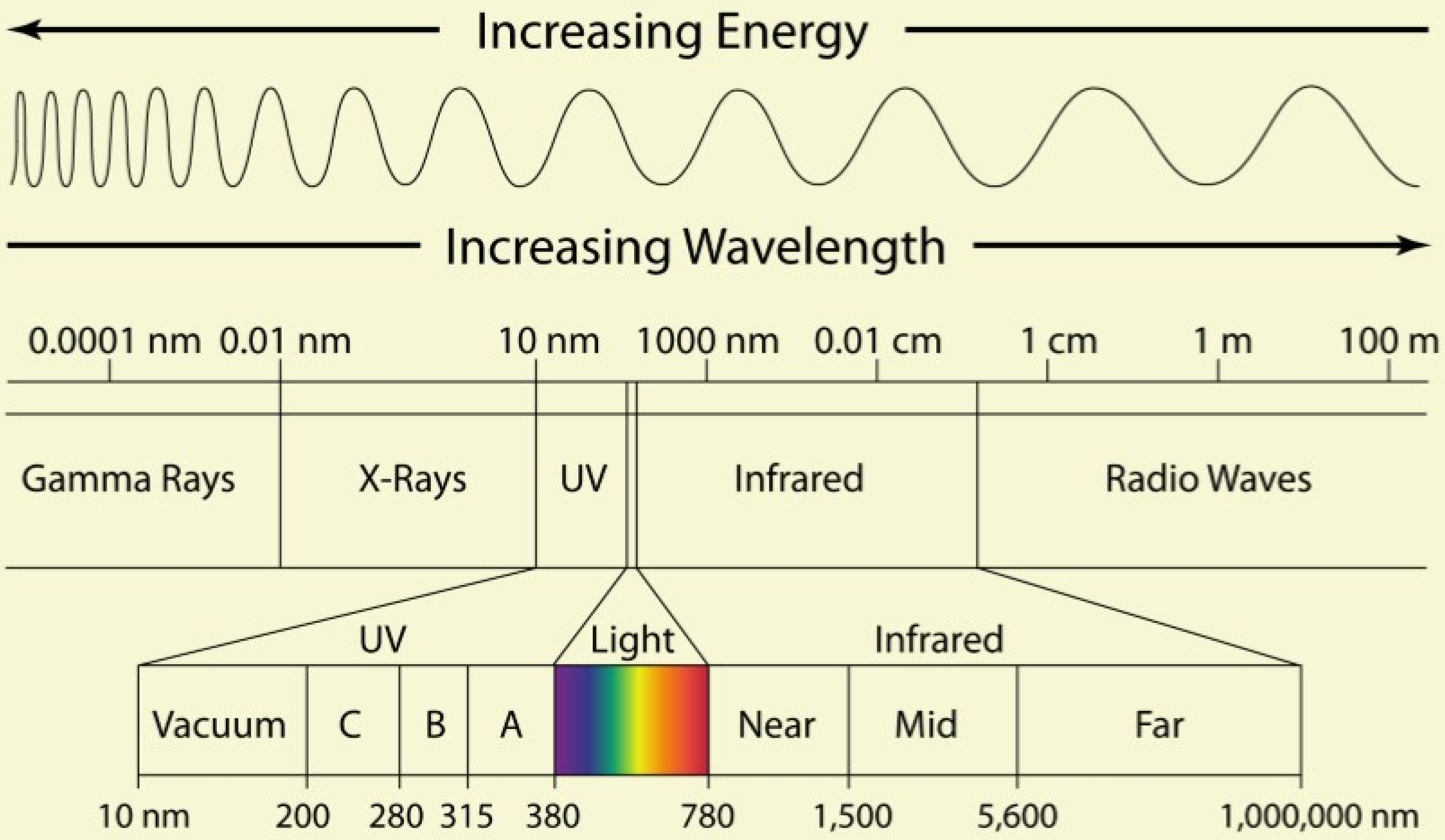
C. 1 & 2 only

D. Neither 1 nor 2

STRUCTURE OF THE ATMOSPHERE

4 THERMOSPHERE

IONOSPHERE



EXPLANATION

- Statement 1 is wrong as the presence of ionosphere causes the reflection of radio waves to Earth and not ozone.
- Statement 2 is correct. As can be seen in picture, radio waves have long wavelengths.

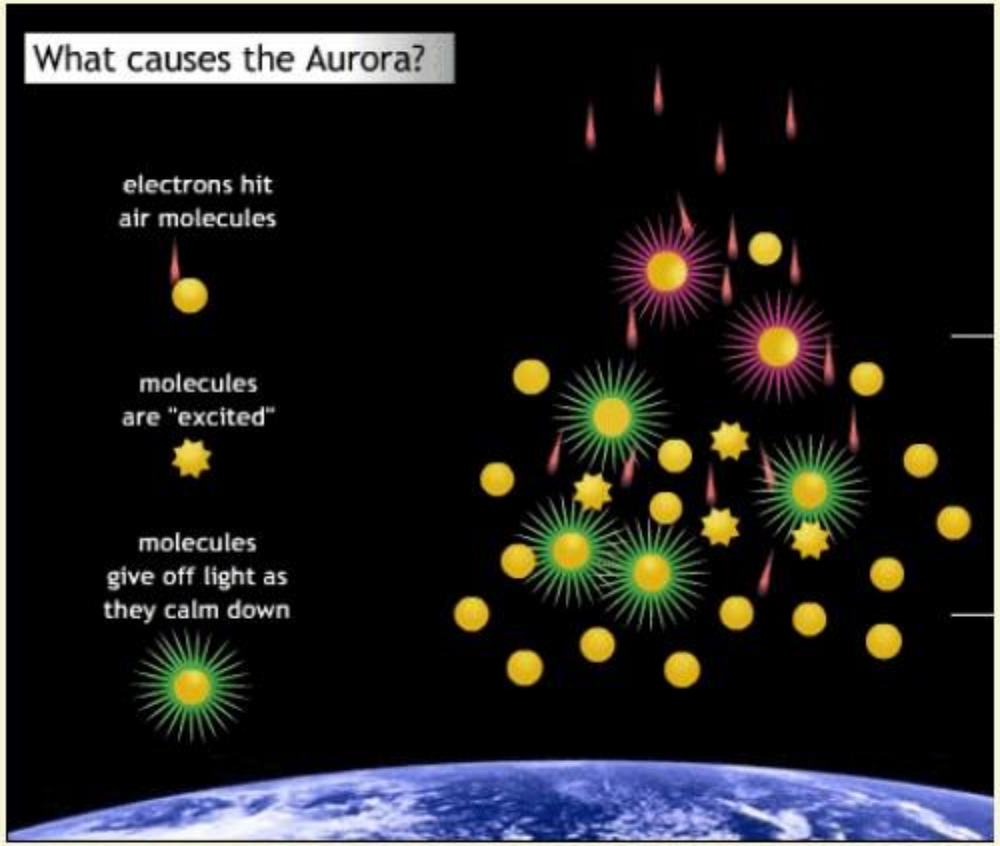
Learning Space

STRUCTURE OF THE ATMOSPHERE

4 THERMOSPHERE

IONOSPHERE

AURORA AUSTRALIS AND AURORA BOREALIS



1. The Aurora Australis and Aurora Borealis are observed in the Ionosphere region.

2. The Aurora Australis and Aurora Borealis are cosmic glowing lights produced by a stream of electrons discharged from the sun's surface due to magnetic storms (Coronal Mass Ejections)

WE EXPRESS OUR SINCERE THANKS FOR VIEWING THIS VIDEO

Presented by

Learning Space

For Suggestions:
suggestions@learningspace.in

To Contact us:
info@learningspace.in

OUR TEAM

G. V. Rao	Amrita Naidu
K. Srikanth	M. Binukrishna
D. Sunil Kumar	G. Ravi Babu
L. V. Krishna	D. Joji
D. Chaitanya	K. Victor Babu

Visit us at:
www.learningspacedigital.com
0866-24 44472
098499 42299